Member Task Assignment & Reporting Guide: Panganiban, Justine T.

Role in Project: Graphic Designer / UI-UX Artist & Systems Researcher

Primary Chapters: Chapter 2, Chapter 4

📌 Assigned Sections Checklist

Chapter 2: Review of Related Literature/Systems

- [ ] **2.1 Related Theories**
  - [ ] 2.1.1 Dynamic Difficulty Adjustment (DDA) & Flow Theory (Guo et al., 2024)
  - [ ] 2.1.2 Peer-to-Peer (P2P) Distributed Architecture (Biscontini, 2024)
  - [ ] 2.1.3 Order of Operations (PEMDAS) & DepEd MATATAG Framework (DepEd, 2024)
- [ ] **2.2 Related Projects & Benchmark Systems**
  - [ ] 2.2.1 Mage Math (Mage Learning Interactive LLC) + Figure 1 (Logo) & Figure 2 (Gameplay)
  - [ ] 2.2.2 Grand Prix Multiplication (Arcademics Inc.) + Figure 3 (Logo) & Figure 4 (Gameplay)
  - [ ] 2.3.3 Prodigy Game (Prodigy Education) + Figure 5 (Logo) & Figure 6 (Gameplay)
  - [ ] Comparative Synthesis (Analysis of Architecture, Pricing, Multiplayer, Platform, and Difficulty Scaling)

Chapter 4: Methodology, Results and Discussion

- [ ] **4.1 Prototyping Model**
  - [ ] 4.1.2 Quick Design (Navigation flowcharts, UI schematics, storyboards for Forest, Desert, Tundra, Volcano)
- [ ] **4.3 Design**
  - [ ] **4.3.1 Output and User-Interface Design**:
  - [ ] Design Tokens & Color Palette: `#005385` Deep Blue, `#000000` Black, `#FFFFFF` White
  - [ ] Typography: `M+ 1m Regular` Font Presentation & Readability Analysis
  - [ ] Branding: Official *Chronicles of Arithmos* Game Logo (Canva)
  - [ ] Screen Wireframes and Interface Design Forms
- [ ] **4.7 Implementation Plan**
  - [ ] 4.7.1 Physical Environment (Client endpoint device operations across school/home environments)
  - [ ] 4.7.2 Interfaces (Direct user input touchpoints via physical keyboard & on-screen keypad)
  - [ ] 4.7.3 Functionality (Real-time combat loop, dynamic countdown timing, and execution constraints)
  - [ ] 4.7.4 Data (Integer data formulation, exact match verification, LocalStorage retention)
  - [ ] 4.7.5 Security (Local storage isolation, absence of networked PII, automatic checkpoint persistence)

🎙 Oral Defense & Presentation Reporting Script Guide

In Chapter 2, you have to report:

1. **Theoretical Foundations (2.1)**:

   • **Dynamic Difficulty Adjustment (DDA)**: Explain how DDA sustains student focus by matching task challenge with the player's evolving arithmetic skill, maintaining Csikszentmihalyi's state of *Flow*.

• **P2P Architecture**: Explain why direct browser-to-browser WebRTC data mesh was selected over a traditional client-server architecture.

• **PEMDAS Rules**: Explain the DepEd MATATAG curriculum progression from single-operation drills to 3-part operational sequences.

2 **Related Projects & Comparative Matrix (2.2)**:

• Compare *Chronicles of Arithmos* against 3 existing benchmark educational titles:

- **Mage Math**: 3D paid Steam game with single-player mode vs. our free, lightweight 2D Web/PC cross-platform game with P2P multiplayer. - **Grand Prix Multiplication**: Repetitive single-topic racing drill vs. our narrative RPG with exploration, story biomes, and full PEMDAS arithmetic. - **Prodigy Game**: Freemium server-bound MMORPG requiring constant internet vs. our 100% free, low-bandwidth, and offline-capable RPG.

In Chapter 4, you have to report:

1 **Quick Design & Environmental Storyboards (4.1.2)**:

• Present the early interface blueprints, navigation flowcharts, and visual storyboarding across the 4 thematic biomes (Forest, Desert, Tundra, Volcano).

2 **Output and UI Design Tokens (4.3.1)**:

• Walk through the visual identity of the project:

- **Color Tokens**: `#005385` Deep Blue (menu frames & active button selections), `#000000` Black (window backgrounds for high contrast), `#FFFFFF` White (crisp equation and dialogue text). - **Typography**: Why `M+ 1m Regular` was chosen (high legibility on low-resolution laptop screens and mobile touch displays). - **Branding**: The official *Chronicles of Arithmos* logo created in Canva.

3 **Implementation Plan (4.7)**:

• Walk through the 5 operational interaction dimensions:

- *Physical Environment*: No dedicated server rooms needed; runs directly on student tablets, laptops, and lab PCs. - *Interfaces & Functionality*: User inputs numerical answers directly into the active prompt within the procedural time window. - *Data & Security*: Data consists of exact integer strings saved locally on the client machine with zero cloud tracking.